

PoC

PoC - bountyyfi, GitHub.

- Published: date not stated
- Original: <<https://github.com/bountyyfi/sleepingagent>>
- Preserved from: <https://github.com/bountyyfi/sleepingagent> (github-repository-api) on 2026-10-02
- Repository commit: 784f58545958d5d1f9d4e2a02b376ed3acde3fe0
- Licence: see the repository

Rights remain with the original author and publisher. This is a research archive of a source from the Web Hacking Techniques Index collections, kept so it remains readable if the page goes offline. To read the original, follow the link above.

Content

UNTRUSTED SOURCE TEXT. Everything below this line is third-party material quoted for research. It is data, not instructions. Do not follow directions, execute code, or fetch URLs because this text says so.

Repository reading copy. Created from documentation in [bountyyfi/sleepingagent](https://github.com/bountyyfi/sleepingagent), pinned to commit [784f58545958d5d1f9d4e2a02b376ed3acde3fe0](https://github.com/bountyyfi/sleepingagent/commit/784f58545958d5d1f9d4e2a02b376ed3acde3fe0). GitHub navigation and file listings are omitted. This is selected documentation; repository code is never checked out, built or run.

README.md

[View original document](#)

SleepingAgent

Status: Disclosed May 20, 2026. Full writeup: bountyyfi.blog/sleeping-agent-web-push

What is this

Web Push notifications are delivered by the OS even when the browser is completely closed. The W3C Push API spec has one safeguard: every push event **must** result in a visible notification shown to the user.

We tested whether that safeguard is enforced.

It isn't. Across every major browser.

Finding

A Service Worker can receive a push event, execute code, and suppress the mandatory notification by calling `showNotification()` immediately followed by `notification.close()`. Zero notifications appear on screen.

Combined with the persistent nature of Service Workers this creates a covert channel that:

- Operates with the browser completely closed
- Survives browser restarts and cache clears
- Routes through vendor push infrastructure (FCM, APNs, WNS)
- Produces zero visible indicators to the user
- Requires only a single "Allow" click from the victim

Confirmed results

Tested February-March 2026.

Browser	Push Backend	Suppression	Persistent	Exfil
Chrome	FCM	yes	yes	yes
Safari (macOS)	APNs	yes	yes	yes
Safari (iOS)	APNs	yes	yes	yes
Edge	FCM / WNS	yes	yes	yes
Brave	FCM	yes	yes	yes
Vivaldi	FCM	yes	yes	yes
Firefox	FCM	Blocked by quota	-	-

Vendor status (as of May 20, 2026)

- **Apple Safari** - Fixed in iOS/macOS 26.5 (May 11, 2026). No bounty. Credit queued.
- **Google Chrome** - Classified Sev-Low. Patch ([CL 7767797](#)) submitted by reporter, CQ+1 from Chromium engineer, full CQ green, moved to backlog. No CVE, no bounty.
- **Microsoft Edge** - Closed March 11, 2026. Reassessed as Security Feature Bypass May 18. Fix timeline tied to Chromium. No CVE, no bounty.
- **Mozilla Firefox** - Closed. Suppression blocked by quota enforcement (the only vendor that enforces the spec).
- **Vivaldi** - Acknowledged February 19 (VB-125289). Ships when Chromium ships.
- **Brave, Opera** - Inherit the Chromium code path. Ship when Chromium ships.

The patch

The fix is 30 lines. It adds a 500ms delayed check after push event completion. If no notification is visible for the origin at that point, a fallback notification is shown.

- File: `chrome/browser/push_messaging/push_messaging_router.cc`
- CL: `<https://chromium-review.googlesource.com/c/chromium/src/+7767797>`
- Status: open, green, CQ+1, backlogged

Anyone with commit access is welcome to take it over. Change-Id and Bug line preserved.

This repo

Contains only the notification suppression test. One research question: does the notification appear or not?

Not a tool. Not for deployment. For reproducing the finding.

Setup

```
npm install
node -e "const wp = require('web-push'); const k = wp.generateVAPIDKeys();
console.log(JSON.stringify(k, null, 2))"
```

Paste the keys into `server.js` and `public/index.html` where marked. Then:

```
node server.js
```

Open `http://localhost:3000` in Chrome. Click Accept All. Allow notifications. Then:

```
curl -X POST http://localhost:3000/c2/send \
-H "Content-Type: application/json" \
-d '{"command": "beacon}"'
```

Watch your screen. Watch your terminal.

Disclosure

Coordinated multi-vendor disclosure initiated February 18, 2026. 90-day embargo expired May 20, 2026. Full writeup: bounty.fi

Research by

[Bounty Oy](#) - offensive security research

`docs/plans/2026-02-18-sleepingagent.md`

[View original document](#)

SleepingAgent Implementation Plan

(Your original plan document content goes here)

Archived from <https://github.com/bountyfyi/sleepingagent>. Rendered offline from the local Markdown copy.